

Context Surgeon

Explainable context compression for LLM requests

First MVP submission | 09 October 2026

01 PROTOTYPE DESCRIPTION

A lightweight local API and browser interface that accept a system prompt, reference context and user query, then return a shorter, query-relevant context. The MVP aims to reduce unnecessary input tokens while retaining the evidence needed to answer. Compression runs locally without an extra LLM call.

02 KEY FEATURES & FUNCTIONALITY

Explainable compression. Splits content into blocks, scores relevance, removes exact duplicates and explains why each block was kept or removed.

Evidence protection. Preserves system instructions, corrections, constraints, recent blocks, linked identifiers and Python dependencies. A soft budget allows essential evidence to exceed the target.

Evaluation workflow. Includes 40 labeled synthetic cases across conversations, documents, code and structured data; local benchmarking; and downloadable JSON reports.

Automatic Gemini selection. Finds a compatible model, switches on model availability or quota errors, and uses the same model for both answers in each comparison. Successful API stages are cached for resume.

Tools: Python 3.10+ standard library (HTTP server, AST, JSON, urllib); HTML, CSS and JavaScript; Google Gemini API for answer comparisons and actual token counts. No third-party packages are required to run the app.

40

Synthetic benchmark cases

29.61%

Mean estimated token reduction

40/40

Cases retaining labeled evidence

Local counts estimate UTF-8 bytes / 4. Labeled-evidence retention is not answer correctness; manual review of generated answers remains pending.

03 TEAM DETAILS

Saad

Dataset preparation

Tarun

Ideation

Gagan

Design

04 PROTOTYPE DEMONSTRATION

Local demo: <http://127.0.0.1:8000> | Start with `python app.py`

Select a sample or paste input, run **Compress locally**, inspect section decisions, then run the offline benchmark or **Compare this example**. The server chooses the Gemini model automatically.

The demo URL works on the computer running the app; this MVP has no public deployment. Gemini evaluation requires an API key and available quota.

05 MVP SCOPE & NEXT STEPS

This is the first working MVP. Windows file-save retries and recovery preserve completed evaluation stages. Further work includes broader robustness testing, semantic relevance, larger datasets and formal human scoring of answer quality.